

SABYASACHI GHOSH

Senior SDET • AI QA Lead • Test Automation Architect

Greater Noida, India • sabyaqa@gmail.com • +91-9650528253

github.com/sabyag42 • sabyag42.github.io • [LinkedIn](#)

PROFESSIONAL SUMMARY

Senior SDET and AI QA Lead with 11+ years across telecoms, HCM, fintech, and banking — combining deep test automation expertise with hands-on AI engineering. At Vonage (Ericsson Group), I built cross-selling API platforms from scratch, automated full microservice test suites running on Kubernetes, and embedded AI-assisted testing tools into daily engineering workflows. I build frameworks that are production-ready, not proof-of-concept — using Python, Playwright, TypeScript, and AI tooling to deliver quality at scale. Certified in LLM evaluation and AI/ML testing.

CORE SKILLS & TOOLS

Languages	Python(Primary),Typescript(Playwright based),Java(Previous stack), Bash
Test Frameworks	pytest, requests API Testing,Playwright, Selenium, Rest Assured,TestNG, Maven, cucumber,K6,Locust for performance
AI & LLM Tools	DeepEval, RAGAS, LangSmith, Langfuse, LangChain, LangGraph, CrewAI, FastMCP
AI Coding Assistants	OpenCode, AWS Kiro, Cursor, GitHub Copilot, Glean Agent
Cloud & Infra	AWS (ECS, S3, DynamoDB, Bedrock, IAM), GCP(Google cloud platform), Docker, Kubernetes, Helm
Observability	Grafana, Prometheus, OpenSearch
Databases	MongoDB, Oracle, DynamoDB
CI / CD	GitHub Actions, Jenkins
Other Tools	Jira, Allure, K6, UFT/VBScript

PROFESSIONAL EXPERIENCE

Vonage — An Ericsson Company | Greater Noida, India

Senior SDET & Gen AI QA Lead Aug 2022 – Present

- ▶ Built the cross-selling API platform from scratch — designed and automated the full test suite covering billing, pricing, routing, and rating microservices (vBilling, vPricing, vRouter, vRater) and drove it to production. Maintained the complete test layer and performance coverage across all services.
- ▶ Owned the Sigma UI (pricing UI framework) test automation end-to-end using Playwright with TypeScript — covering functional, regression, and edge-case flows in a component-heavy frontend.
- ▶ Led quality for the Prepaid USD amplification feature from day one — involved from initial design through production rollout. This feature doubled prepaid growth for the organisation.
- ▶ Designed and deployed a complete UI and API automation framework in Python (pytest) for microservices running in an AWS ECS cluster. Tests are executed by spinning up Kubernetes pods via Helm charts, enabling isolated and reproducible test runs in production-like environments.
- ▶ Built MongoDB test helpers and integrated them into the framework to validate data state across services — covering write paths, aggregation, and edge conditions that purely API-level checks would miss.
- ▶ Integrated AI-assisted tooling across the engineering workflow: OpenCode and AWS Kiro for code generation and test scaffolding; Cursor and GitHub Copilot for code review augmentation; Glean Agent for knowledge retrieval. Connected local LLM flows with AWS Bedrock for test intelligence use cases.

- ▶ Built a DeepEval-based AI testing framework for end-to-end LLM quality evaluation — covering hallucination detection, context precision, and answer relevancy for AI-powered product features.
- ▶ Set up performance monitoring with Grafana, Prometheus, and OpenSearch dashboards — giving the team real-time visibility into test health and service behaviour under load (K6).

UKG (Ultimate Kronos Group) | Noida, India

Senior SDET Jan 2019 – Jul 2022

- ▶ Led the end-to-end test framework build for the Workforce Management (Attendance) product — covering UI automation with Python/pytest and Selenium, and API layer testing from scratch for the IA team.
- ▶ Key contributor to the IP Restriction security feature — designed and executed test scenarios covering whitelisted/non-whitelisted IP access, boundary conditions, and enforcement of access policies across customer-facing application flows.
- ▶ Ran load and performance tests with JMeter to validate system behaviour under concurrent user sessions during the IP restriction rollout.
 - Stack: Python, pytest, Selenium, Java (selective), JMeter, GCP, Jira

Tata Consultancy Services (TCS) | Gurugram, India

Senior Test Engineer — Client: American Express Mar 2018 – Dec 2019

- ▶ Owned end-to-end testing for the American Express AI chatbot — the same chatbot visible on their website today. Designed the test framework, built the test suite, and validated conversational flows, NLU accuracy, and fallback behaviour from scratch.
- ▶ Automated Oracle application testing using Selenium with Java (TestNG, Maven) — covering critical backend data validation flows integrated with the chatbot platform.
- ▶ Set up the Jenkins CI pipeline for continuous test execution — enabling daily regression runs without manual intervention.
 - Stack: Selenium, Java, TestNG, Maven, Jenkins, Oracle, AI chatbot testing frameworks

DXC Technology (onsite UK) | United Kingdom (Onsite)

QA Engineer Jun 2014 – Feb 2018

- ▶ Delivered testing and automation for major UK banking and insurance clients including TSB, Santander, Barclays, and Aviva — building and implementing test frameworks that the clients adopted at scale.
- ▶ Designed and deployed Selenium + UFT (VBScript) frameworks across client environments — reducing manual testing cycles and earning direct commendations from client leads.
- ▶ Worked onsite in the UK managing client relationships and delivery expectations across multiple concurrent banking programmes.
 - Stack: Selenium, Java, Maven, TestNG, Jenkins, UFT/VBScript, AWS

PERSONAL PROJECTS & OPEN SOURCE

selenium-ai-framework | Python, Pytest, GPT-4o-mini, Allure, Kubernetes

- ▶ Production-grade Selenium 4 + pytest framework using uv for dependency management. Integrates GPT-4o-mini for AI-assisted failure analysis and self-healing locators. Structured logging via structlog, Allure reporting, targeting rahulshettyacademy.com practice flows.

mmt-playwright-ts | TypeScript, Playwright, GPT-4o

- ▶ 7-layer enterprise Playwright TypeScript framework for MakeMyTrip — includes self-healing locators, AI failure analysis using GPT-4o, and GitHub Actions CI. Designed as a portfolio-grade production framework.

ai-test-framework-k8s | FastAPI, GPT-4o, Kubernetes, Helm, GitHub Actions

- ▶ AI-powered test framework deployed on Kubernetes — FastAPI backend, GPT-4o for test case generation from Jira tickets, Helm charts for deployment, HPA for scaling, GitHub Actions for CI/CD. Live at github.com/sabyag42/ai-test-framework-k8s.

crewai-research-crew | Python, CrewAI, LangChain

- ▶ 3-agent sequential AI pipeline (Researcher → Analyst → Writer) using CrewAI. Demonstrates multi-agent orchestration, inter-agent communication, and structured output delivery.

langchain-mastery monorepo | Python, LangChain, LangGraph, RAG, FastMCP, LangSmith

- ▶ End-to-end LLM engineering monorepo covering LangChain, LangGraph agents, RAG pipelines, and FastMCP server integration (connected to Claude Desktop). Full LangSmith tracing configured across all modules.

Resume Tailor Web App | *React, Vite, GPT-4o, PDF Export*

- ▶ React-based web app that takes a job description and base resume and generates a tailored resume using GPT-4o. Includes PDF export. Deployed to GitHub Pages.

Hello, Sparky! — Children's AI Book | *Amazon KDP*

- ▶ Published children's book on AI concepts — illustrated DOCX and ebook PDF, live on Amazon KDP. Makes AI accessible for young readers.

EDUCATION & CERTIFICATIONS

- ▶ B.Tech / B.E. in Electrical and Electronics — West Bengal University of Technology, 2014
 - ▶ ISTQB Certified Tester — Foundation Level
 - ▶ O'Reilly AI/LLM Testing Certification (2025–2026)
 - ▶ DeepLearning.AI — LLM Evaluation and Observability (2026)
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